

MENTORA

Multi-Tenant AI-Powered Learning Experience Platform

Product Requirements Document

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Change in v3.0	Added Sub-Admin (Department Admin) role

C O N F I D E N T I A L

DOCUMENT REVISION HISTORY

Version	Date	Author	Changes
1.0	Jan 10, 2025	Product Team	Initial draft
2.0	Jan 15, 2025	Product Team	MongoDB architecture, user provisioning methods
3.0	May 2025	Product Team	Added Sub-Admin (Department Admin) role — tenant-created middle-management layer for instructor & course oversight

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1. EXECUTIVE SUMMARY

1.1 Product Overview

Mentora is an enterprise-grade, multi-tenant Software-as-a-Service platform designed to revolutionize institutional learning through AI-enhanced education delivery. The platform enables universities, corporations, and educational institutions to deploy a fully-branded, intelligent learning ecosystem in under 60 minutes without any infrastructure overhead.

1.2 Problem Statement

Educational institutions currently face three critical challenges:

Fragmented Technology Stack

- Universities use 5–8 different tools (Zoom, Google Classroom, WhatsApp, separate LMS) with no integration
- Data silos prevent unified analytics and reporting
- Instructors spend 40% of time on tool management rather than teaching

Zero AI Integration

- Traditional LMS platforms offer no intelligent tutoring
- Students struggle after class hours with no 24/7 doubt support
- No predictive analytics to identify at-risk students early

Administrative Overload

- Tenant Admins (institution administrators) are overloaded managing hundreds of instructors, courses, and departments
- No middle management layer between the institution admin and ground-level instructors
- Traditional LMS deployment takes 6–8 months with expensive licensing (15–30 USD per student annually)

1.3 Solution

Mentora provides a unified, AI-native platform with a four-tier user hierarchy that distributes administrative load intelligently:

Super Admin (Mentora Team)	Tenant Admin (Institution)	Sub-Admin (Dept. Manager) ★	End Users (Instructor / Student)
• Platform-wide control • Tenant provisioning • Billing & monitoring • Security & updates	• Institution branding • Create Sub-Admins ★ • Bulk user import • Institution analytics	• Manage assigned instructors ★ • Oversee dept. courses ★ • Dept. analytics ★ • Cannot access other depts. ★	• Create & teach courses • Enroll & learn • AI tutoring 24/7 • Track own progress

1.4 Business Model

Revenue Model: SaaS subscription with three tiers

Tier	Target Market	Pricing	Min. Commitment
Starter	Coaching institutes (<500 students)	15 INR/student/month	7,500 INR/month
Professional	Private colleges (500–5,000 students)	12 INR/student/month	Annual contract
Enterprise	Universities & corporations (5,000+ learners)	8–10 INR/student/month	Custom negotiation

Additional Revenue Streams

- Setup fees (25,000–100,000 INR one-time)
- Blockchain certificates (50 INR per certificate)
- Premium support packages (50,000 INR monthly)
- Custom mobile app development (500,000 INR one-time)
- API access and integrations (included in Enterprise)

1.5 Market Opportunity

Year	Institutions	Students	ARR
Year 1	100	200,000	30 Crore
Year 3	500	1.5 million	180 Crore
Year 5	2,000	6 million	720 Crore

2. PRODUCT VISION & STRATEGIC POSITIONING

2.1 Vision Statement

Our Vision

- To become the operating system for institutional learning in India and emerging markets — where every educational interaction is enhanced by AI, every student receives personalized mentorship at scale, and every institution can compete with world-class universities through technology democratization.

2.2 Mission Statement

Democratize access to high-quality, AI-enhanced education by providing institutions with enterprise-grade learning infrastructure at a fraction of traditional costs, while maintaining the highest standards of data security, privacy, and pedagogical effectiveness.

2.3 Core Values

Education First

- Every product decision prioritizes learning outcomes over engagement metrics
- AI assists and augments human teaching, never replaces it

Data Sovereignty

- Institutions own 100% of their data
- Easy export and migration capabilities built from day one

Inclusive Access

- Support for 12+ Indian and international languages
- WCAG 2.1 Level AA accessibility compliance
- Works on low-bandwidth connections (2G/3G optimization)

2.4 Product Positioning

For educational institutions that need a complete, AI-powered learning platform with intelligent administrative delegation, Mentora is a multi-tenant SaaS solution that deploys in under an hour and provides enterprise-grade features — including a Sub-Admin layer that distributes institutional management load — at one-third the cost of traditional LMS platforms.

2.5 Competitive Differentiation Matrix

Dimension	Mentora	Canvas/Blackboard	Video Tools	B2C EdTech
Deployment Speed	Under 1 hour	6–8 months	Instant but limited	2–4 weeks
AI Integration	Native, contextual	None or basic	None	Generic
Sub-Admin Layer	Yes — tenant-created ★	No	No	No
White Labeling	Full (domain + apps)	Limited	None	None

Pricing	12–15 INR/student/month	300–600 INR	450 INR/user	1,200–2,400 INR
Data Ownership	Institution owns all	Institution owns	Vendor retention	Vendor retention

3. MARKET ANALYSIS & OPPORTUNITY

3.1 Market Size & Growth

Market Segment	Key Metrics
Indian EdTech Market (2024)	Current: 750M USD Projected 2028: 4B USD CAGR: 39.77%
Higher Education	50,000+ institutions 38M students 1.5M faculty 70% still offline
Corporate L&D	50,000 Crore annual spend 85% outsourced training
Coaching & Test Prep	15M students enrolled 35,000 Crore market 40% now hybrid/online

3.2 Target Customer Segments

Primary: Private Colleges (Tier 2/3 Cities)

- 500–5,000 students; limited IT budget; NAAC/NBA accreditation pressure
- Pain: Cannot afford expensive LMS; faculty resistance to multiple disconnected tools
- Value: 12 INR/student/month; Sub-Admin layer reduces admin workload across departments

Secondary: Corporate Training Academies

- 1,000–10,000 employees; compliance tracking; HRMS integration needed
- Value: Build internal university; AI tutoring reduces live instructor dependency

Tertiary: Premium Coaching Institutes

- 5,000–50,000 students; star-instructor dependency risk
- Value: Record best lectures once; AI handles 70% of routine doubts

3.3 Market Entry Phases

Phase	Period	Target	Success Metric
Beachhead	Months 1–6	10 private colleges in Rajasthan	80% pilot-to-paid conversion
Expansion	Months 7–12	40 institutions across 3 segments	50 institutions, 30 Crore ARR
Scale	Year 2–3	National + corporate + international	500 institutions, 180 Crore ARR

4. SYSTEM ARCHITECTURE OVERVIEW

4.1 Architecture Philosophy

Mentora operates on a Single Codebase, Multi-Tenant architecture. One application instance serves all clients while maintaining complete logical and security isolation between organizations. Data is strictly partitioned using tenant identifiers embedded in every database record.

4.2 Four-Tier User Hierarchy

★ **NEW** Version 3.0 introduces a fourth tier — the Sub-Admin (Department Admin) — created and managed by the Tenant Admin to distribute institutional load.

#	Tier	Who	Core Responsibility
1	Super Admin	Mentora Developers (you)	Platform-wide control: tenant provisioning, billing, monitoring, deployments, security
2	Tenant Admin	Institution / Organization (your client)	Full institution management: branding, user import, Sub-Admin creation, institution-wide analytics
3	Sub-Admin ★	Department Manager (created by Tenant Admin)	Department-scoped management: instructor oversight, course approval, department analytics
4	End Users	Instructors, Students, TAs, Parents	Teaching, learning, grading — fully scoped to their courses within their institution

4.3 Data Isolation Strategy

Layer	Mechanism	Protection
Layer 1 — Database	MongoDB: tenant_id on every record; compound indexes lead with tenant_id	Middleware injects filter on all queries; writes fail without tenant_id
Layer 2 — Application	Request context middleware validates token + tenant on every HTTP request	Rate limiting & resource quotas enforced per tenant
Layer 3 — Storage	S3 prefix: /tenant_id/file_category/file_id	Pre-signed URLs with tenant validation; CORS restrictions
Layer 4 — Vector DB	Dedicated Pinecone namespace per tenant	Cross-namespace queries architecturally impossible
Layer 5 — Sub-Admin ★	department_id filter applied on top of tenant_id for all Sub-Admin queries	Sub-Admin cannot query instructors/courses outside their assigned department

4.4 Technology Stack Summary

- Frontend: Next.js 14 · Tailwind CSS · shadcn/ui · Zustand · React Native (mobile)
- Backend: FastAPI (Python) · JWT RS256 · Kong API Gateway · Celery/Redis workers
- Database: MongoDB Atlas (Mumbai primary, Singapore replica) · Redis cache · Elasticsearch
- Storage: AWS S3 + CloudFront CDN · AWS MediaConvert (HLS streaming)
- AI: Llama 3.1 70B (local) + GPT-4o (cloud) · Pinecone vector DB · Whisper speech-to-text

- Live Classes: Agora.io WebRTC · up to 500 participants · auto-recording
- DevOps: Docker + Kubernetes (AWS EKS) · GitHub Actions CI/CD · Terraform IaC · Datadog + PagerDuty

4A. ROLE HIERARCHY — FULL PLATFORM MAP

★ **NEW** This section is updated in v3.0 to include the Sub-Admin tier with full permission boundaries.

Complete Permission Matrix

Permission	Super Admin	Tenant Admin	Sub-Admin ★	Instructor	Student
Create / delete tenants	✓	✗	✗	✗	✗
View all tenants	✓	✗	✗	✗	✗
Manage platform billing	✓	✗	✗	✗	✗
Deploy platform updates	✓	✗	✗	✗	✗
Configure institution branding	✓	✓	✗	✗	✗
Create / manage Sub-Admins	✓	✓	✗	✗	✗
Assign instructors to Sub-Admin	✓	✓	✗	✗	✗
Bulk import users (institution-wide)	✓	✓	✗	✗	✗
View institution-wide analytics	✓	✓	✗	✗	✗
View department analytics	✓	✓	✓ (own dept)	✗	✗
View assigned instructors	✓	✓	✓ (own dept)	✗	✗
Approve / reject courses (dept)	✓	✓	✓ (own dept)	✗	✗
Manage course content	✓	✓	✓ (own dept)	✓ (own)	✗
View student list in dept. courses	✓	✓	✓ (own dept)	✓ (enrolled)	✗
Conduct live classes	✗	✗	✗	✓	✗
Submit assignments	✗	✗	✗	✗	✓
Use AI tutor	✗	✗	✗	✗	✓
View own grades	✗	✗	✗	✗	✓

Visibility Boundaries

What Super Admin Can See

- Every tenant, every user, every piece of content across the entire platform
- All billing data, resource consumption, AI token usage per tenant
- Infrastructure health, server logs, security events

What Tenant Admin Can See

- All users, courses, and data within their institution only
- All Sub-Admins they have created and their department assignments
- Institution-wide analytics, attendance, assessment scores
- Cannot see any other tenant or Mentora internal operations

What Sub-Admin Can See (New in v3.0)

- Only the instructors assigned to their department by the Tenant Admin
- Only the courses created by instructors in their department
- Only the students enrolled in courses within their department
- Department-scoped analytics only — no cross-department visibility
- Cannot create other Sub-Admins, cannot modify branding, cannot access billing

What Instructors Can See

- Only the courses they personally teach within the institution
- Only the students enrolled in their courses
- Their own performance analytics and grading queue

What Students Can See

- Only courses they are enrolled in within their institution
- Their own grades, study plan, AI chat history
- Discussion forums for their enrolled courses only

5. FEATURE SPECIFICATIONS

5.1 Core Platform Features

Feature 1: Instant Tenant Provisioning

Complete, isolated institutional workspace created in under 60 minutes with zero manual configuration.

Functional Requirements

Requirement	Description
FR-1.1 One-Click Creation	Auto-generates tenant ID, subdomain, SSL certificate within 5 minutes
FR-1.2 Infrastructure Setup	MongoDB namespace, Pinecone namespace, S3 bucket, default configs auto-created
FR-1.3 Admin Initialization	First admin account created; 24-char temp password; forced reset on first login
FR-1.4 Setup Wizard	Guided branding, timezone, language, academic calendar, and Sub-Admin role config on first login

Acceptance Criteria

- Provisioning completes in under 10 minutes
- Admin welcome email within 2 minutes
- Auto-rollback if any step fails

Feature 2: User Provisioning and Onboarding

Multiple methods for administrators to create and activate user accounts across all roles.

Functional Requirements

Requirement	Description
FR-2.1 CSV Bulk Upload	Up to 50 MB / ~100,000 rows; validation preview; error report download; batch processing
FR-2.2 Magic Link Activation	48-hour JWT token; branded activation page; real-time password strength meter
FR-2.3 Temporary Password	System-generated 12-char password; forced change on first login
FR-2.4 SSO Integration (Enterprise)	Google, Microsoft 365, Okta, SAML 2.0; JIT provisioning; multi-provider support
FR-2.5 Sub-Admin Provisioning ★	Tenant Admin creates Sub-Admin accounts and assigns departments — see Feature 10 for full spec

Acceptance Criteria

- 10,000 CSV records processed in under 5 minutes
- Magic link emails within 2 minutes
- SSO latency under 3 seconds

Feature 3: AI Teaching Assistant

24/7 intelligent chatbot trained on course materials with Socratic tutoring and multi-language support.

Functional Requirements

Requirement	Description
FR-3.1 Contextual Q&A	Retrieves lecture transcripts, PDFs; cites sources with timestamps; zero cross-tenant leakage
FR-3.2 Socratic Tutoring	Detects homework questions; guides with hints, not direct answers
FR-3.3 Multi-Language	12 languages; auto-detection; technical terms kept in English
FR-3.4 At-Risk Alerts	Alerts instructor after 3+ struggle indicators; aggregate confusion dashboard for Sub-Admin ★
FR-3.5 Rate Limiting	Starter: 50/day · Professional: 200/day · Enterprise: unlimited (20/min)

Acceptance Criteria

- >85% accuracy rate
- Response <5 s (p95)
- Socratic mode >90% of homework questions
- >99.5% uptime

Feature 4: Live Class Studio

Full live teaching environment: video, screen sharing, whiteboard, polls, breakout rooms, auto-recording.

Functional Requirements

Requirement	Description
FR-4.1 Scheduling	Recurring classes; iCal; Google/Outlook sync; capacity & waitlist management
FR-4.2 Session Features	Video gallery (9 feeds); annotations; infinite whiteboard; polls/quizzes; Q&A with upvoting
FR-4.3 Breakout Rooms	2–20 rooms; random/manual/self-select; instructor broadcasts to all rooms
FR-4.4 Recording	Auto-recording → Whisper transcription → speaker diarization → timestamp-synced player
FR-4.5 Attendance	Join/leave timestamps; 75% duration threshold; auto-alerts for <70% attendance

Acceptance Criteria

- Join success >99%
- Audio/video latency <300 ms (p95)
- 500 concurrent without degradation
- Recording available within 30 min

Feature 5: Personalized Learning Paths

AI-generated day-by-day study plans adapting to each student's performance and deadlines.

Functional Requirements

Requirement	Description
FR-5.1 Performance Analysis	Tracks quiz scores, video watch time, AI queries; identifies weak/strong areas
FR-5.2 Study Plan Generation	Spaced repetition, interleaving, retrieval practice; specific daily schedule
FR-5.3 Dynamic Adjustment	Replanning within 24 hours of new data; handles surprise deadlines
FR-5.4 Multi-Course Balancing	Holistic view across 4–6 courses; max 20 hours/week recommended
FR-5.5 Progress Visualization	Course health meters; weekly adherence %; projected final grade

Acceptance Criteria

- Average 12% grade improvement over 8 weeks
- Plan adherence >60%
- Regeneration within 10 seconds

Feature 6: Real-Time Cornell Notes Generation

AI generates structured Cornell-format notes synchronized to lecture timeline during live classes.

Functional Requirements

Requirement	Description
FR-6.1 Live Transcription	Real-time Whisper; speaker identification; multi-language code-switching
FR-6.2 Cornell Format	Cues column (questions/formulas), Notes column (content), Summary section
FR-6.3 Timestamp Sync	Click note item to jump to that video moment; personal annotations
FR-6.4 Export	PDF, Notion, Evernote, Google Docs; print-friendly format

Acceptance Criteria

- Notes within 5 min of lecture end
- >80% of concepts captured
- >4/5 student usefulness rating

Feature 7: AI-Powered Grading Assistant

Auto-grading for objective questions; AI-assisted scoring for subjective responses — reduces grading time 60%.

Functional Requirements

Requirement	Description
FR-7.1 Objective Auto-Grade	MCQ, multi-select, T/F, fill-in-blank, numerical with tolerance; instant feedback
FR-7.2 Subjective Assist	AI scores per rubric criterion; side-by-side instructor review; one-click accept/adjust
FR-7.3 Code Grading	Docker sandboxed execution; test cases; static analysis; AI code review
FR-7.4 Plagiarism Detection	Turnitin API + GPTZero/Originality.AI; configurable similarity threshold

Acceptance Criteria

- 200 objective assignments graded in <2 min
- AI agrees with instructor within 10% for >80% of submissions

Feature 8: Institutional Knowledge Graph

Semantic network connecting concepts, lectures, courses, and people for cross-course discovery.

Functional Requirements

Requirement	Description
FR-8.1 Entity Extraction	NLP extracts concepts, people, events from all transcripts and PDFs; stored in Neo4j
FR-8.2 Cross-Course Search	"Find all lectures on climate policy" — timeline view with timestamp click-through
FR-8.3 Curriculum Gap Analysis	Compare institution offerings to benchmark (MIT CS); identify missing topics
FR-8.4 Research Export	Anonymized dataset: concept co-occurrence, attention patterns, peer learning effects

Acceptance Criteria

- Search across 10,000 lectures in <3 s
- >85% entity extraction precision

Feature 9: Offline Content Access (Mobile)

Students download videos, PDFs, and notes for offline study without internet connection.

Functional Requirements

Requirement	Description
FR-9.1 Download Management	Per-item and batch download; WiFi-only mode; storage management dashboard
FR-9.2 Offline Viewing	Video player (0.5×–2× speed); PDF viewer with search, bookmarks,

	annotations
FR-9.3 Sync	Auto-uploads watch progress, annotations, quiz answers on reconnect; delta sync only

Acceptance Criteria

- Video files 40% smaller without quality loss
 - Up to 10 GB on-device storage
 - Zero data loss during transitions
-

5A. SUB-ADMIN ROLE — FULL FEATURE SPECIFICATION

★ **NEW** This entire section is new in v3.0. It defines the Sub-Admin (Department Admin) role created and managed by the Tenant Admin.

Overview

The Sub-Admin is a tenant-created middle-management role that sits between the Tenant Admin and the ground-level Instructors and Students. The primary purpose is to distribute institutional administrative load — so the Tenant Admin does not have to personally manage every instructor, course, and department across a large institution. Each Sub-Admin operates strictly within the scope of their assigned department(s).

Design Principle

- A Sub-Admin can never see or touch anything outside their assigned department(s)
- A Sub-Admin is created, configured, and revoked exclusively by the Tenant Admin
- The Sub-Admin role cannot create other Sub-Admins (no recursion)
- All Sub-Admin actions are logged and visible to the Tenant Admin in real time

Feature 10: Sub-Admin Creation & Management

Feature 10: Sub-Admin Creation & Management by Tenant Admin

The Tenant Admin can create Sub-Admin accounts, assign them to specific departments or course groups, and configure exactly what they can access and do.

Functional Requirements

Requirement	Description
FR-10.1 Creating a Sub-Admin	Tenant Admin opens "Team Management" → "Sub-Admins" → "Add Sub-Admin". Enters: full name, email, phone (optional), assigned department(s). System creates account, sends magic-link activation email branded to the institution.
FR-10.2 Department Assignment	Tenant Admin selects one or more departments from a dropdown (e.g., Computer Science, Mechanical Engineering, Sales Training). Sub-Admin's entire scope is limited to these departments. Can be changed at any time.
FR-10.3 Instructor Assignment to Sub-Admin	Tenant Admin picks which instructors fall under each Sub-Admin's supervision. An instructor can only belong to one Sub-Admin's department at a time. Reassignment transfers supervision to the new Sub-Admin.
FR-10.4 Permission Configuration	Tenant Admin toggles what the Sub-Admin can do: Approve courses (on/off), Send announcements to dept. students (on/off), View student grades in dept. (on/off), Export dept. analytics reports (on/off). All other permissions are fixed by the system.
FR-10.5 Sub-Admin Deactivation	Tenant Admin can suspend or permanently delete a Sub-Admin account. Upon deletion: assigned instructors revert to direct Tenant Admin oversight. No data is lost — courses and content remain intact.
FR-10.6 Audit Log Visibility	Every action performed by a Sub-Admin is logged: timestamp, action type, affected resource. Tenant Admin sees the full audit trail in real time. Sub-Admins cannot view or edit their own audit log.

Feature 11: Sub-Admin Dashboard & Instructor Oversight

Feature 11: Sub-Admin Dashboard & Instructor Oversight

The Sub-Admin's primary workspace shows all instructors in their department, their course activity, live class schedules, and engagement metrics.

Functional Requirements

Requirement	Description
FR-11.1 Instructor Overview Panel	Sub-Admin's home dashboard shows a card for each assigned instructor containing: name, photo, number of active courses, next scheduled live class, average student satisfaction rating, and last platform login. Instructors outside the department are invisible.
FR-11.2 Instructor Activity Monitoring	Sub-Admin can click any instructor card to see: list of courses they teach, live class attendance rates, assignment grading turnaround time (avg. hours to grade), student engagement scores per course, and AI query volume from their students.
FR-11.3 Course List by Instructor	Sub-Admin sees a full list of courses organized by instructor. Each course shows: enrollment count, completion rate, average assessment score, last content update, and publication status (Draft / Pending Approval / Published / Archived).
FR-11.4 Live Class Schedule View	Calendar view showing all upcoming live classes scheduled by department instructors. Sub-Admin can see: class title, instructor name, date/time, enrolled student count, and recording status of past classes.
FR-11.5 Student Roster per Course	Sub-Admin can view which students are enrolled in each course within their department (read-only). Can filter by: enrollment date, attendance rate, grade average, AI tutor usage.
FR-11.6 At-Risk Student Alerts	If AI identifies a student as at-risk (low attendance + low quiz scores + high AI query volume on confusion topics), Sub-Admin receives an alert. Can escalate to Tenant Admin or notify the relevant instructor directly from the dashboard.

Feature 12: Sub-Admin Course Management

Feature 12: Course Approval & Management within Department

Sub-Admin manages the course lifecycle within their department — from reviewing drafts to publishing and archiving — removing this burden from the Tenant Admin.

Functional Requirements

Requirement	Description
FR-12.1 Course Approval Workflow	When an instructor submits a course for review, Sub-Admin receives a notification. Sub-Admin reviews: syllabus, learning objectives, content uploads, assessment design. Can Approve (publishes course), Request Changes (sends feedback to instructor with comments), or Reject (with mandatory reason). Tenant Admin can override any Sub-Admin decision.
FR-12.2 Course Editing Permissions	Sub-Admin can edit course metadata (title, description, enrollment settings, schedule). Sub-Admin cannot edit actual course content or assessments —

	only the instructor who owns the course can do that. Sub-Admin can flag content for review by Tenant Admin.
FR-12.3 Course Archiving	Sub-Admin can archive completed courses within their department. Archived courses become read-only for students (they can still view content but cannot submit new work). Students and instructors are notified automatically.
FR-12.4 Announcement Broadcasting	If permitted by Tenant Admin, Sub-Admin can send announcements to all students enrolled in any course in their department. Announcements appear in the notification center and are sent via email. Cannot send to students outside their department.
FR-12.5 Content Moderation Queue	Sub-Admin has access to a moderation queue for their department: flagged discussion forum posts, reported AI responses, copyright-concern flags on uploaded materials. Can approve, remove, or escalate to Tenant Admin.
FR-12.6 Enrollment Management	Sub-Admin can manually enroll or unenroll students in courses within their department (e.g., to handle add/drop requests). Cannot enroll students in courses from other departments. Bulk enrollment from a CSV file supported.

Feature 13: Sub-Admin Analytics & Reporting

Feature 13: Department-Scoped Analytics & Reporting

Sub-Admin has access to a full analytics suite, but strictly scoped to their department — enabling data-driven management without access to institution-wide data.

Functional Requirements

Requirement	Description
FR-13.1 Department Dashboard	Real-time metrics for the department: total students enrolled, average attendance rate across all live classes, average assessment score, AI tutor query volume today vs. last 7 days, content hours delivered this week.
FR-13.2 Instructor Performance Report	Comparative view of all instructors in the department: student satisfaction rating, grading speed (avg. hours), course completion rates, live class frequency. Sub-Admin uses this to identify instructors needing support.
FR-13.3 Student Progress Report	Aggregated student performance data for the department: grade distribution histogram, attendance distribution, at-risk count, top performers. Can drill down to individual student (if Tenant Admin has enabled this permission).
FR-13.4 Course Health Report	For each course: enrollment trend over time, video completion rates, assignment submission rates, quiz score averages, discussion forum activity. Sub-Admin can identify underperforming courses and flag for instructor attention.
FR-13.5 Export Functionality	All analytics reports exportable as PDF or CSV. PDF reports are branded with institution logo and department name. CSV exports are raw data for import into institution's BI tools. Exports are logged in audit trail.
FR-13.6 Accreditation Report Contribution	Sub-Admin can generate department-level data exports in NAAC/NBA-compatible formats. These feed into the Tenant Admin's institution-wide accreditation report — Sub-Admin does not have access to reports from other departments.

Sub-Admin Acceptance Criteria Summary

Area	Acceptance Criterion
Scope Enforcement	Sub-Admin can never retrieve data from outside assigned department(s) — enforced at database query level via compound index (tenant_id + department_id)
Account Creation	Tenant Admin can create a Sub-Admin account and the account receives activation email within 2 minutes
Instructor Visibility	Sub-Admin dashboard loads all assigned instructor cards within 3 seconds, even for departments with 50+ instructors
Course Approval	Course approval/rejection action is logged, instructor is notified within 1 minute, and Tenant Admin can see the decision in their audit trail
Analytics Accuracy	All Sub-Admin analytics metrics match the values shown in Tenant Admin's institution-wide view for the same department
Data Isolation	Zero instances of a Sub-Admin accessing courses, students, or instructors from a different department — validated by automated security tests in CI/CD pipeline
Audit Trail	100% of Sub-Admin actions are logged with: user ID, action type, affected resource ID, timestamp, and IP address

Sub-Admin Non-Functional Requirements

- Dashboard loads within 3 seconds for departments with up to 200 instructors and 10,000 students
- All data visible to Sub-Admin is cached with 60-second refresh; export generation completes in under 30 seconds for reports up to 12 months
- Sub-Admin sessions expire after 60 minutes of inactivity (same as Tenant Admin)
- Sub-Admin login supports MFA (optional, configurable by Tenant Admin per Sub-Admin account)
- Sub-Admin accounts are covered by the same SOC 2 Type II and ISO 27001 controls as all other roles

6. USER EXPERIENCE & WORKFLOWS

6.1 Student User Journey

Phase 1: Account Activation (Day 0)

- Receives branded email → clicks magic link → creates password → tooltip tour (5 steps, skippable)

Phase 2–3: Active Learning (Weeks 1–15)

- Daily: push notification → watch pre-recorded lecture with AI Cornell notes → ask AI tutor → attend live class → complete practice problems
- Weekly: email summary → assignment submission → grade notification → weekly recap stats

Phase 4–5: Exam Prep & Completion

- AI study guide targeting weakest topics → practice exam → spaced repetition flashcards → live review session
- Final grade → downloadable certificate → optional blockchain credential → next-course recommendations

6.2 Instructor User Journey

Course Setup → Teaching Routine → Semester End

- Create course shell → upload materials (auto-transcoded to HLS) → build assessments → submit for Sub-Admin approval ★
- Live class: join 10 min early → teach → polls → breakout rooms → recording auto-saved → attendance auto-logged
- Grading: auto-graded quizzes reviewed → AI grading assistant for essays → publish grades → student notified
- Analytics: identify at-risk students → send personalized outreach → notify Sub-Admin if pattern persists ★

6.3 Tenant Admin User Journey

Onboarding (Day 1)

- 5-step setup wizard: branding → academic calendar → integrations (SSO, SIS) → billing → create Sub-Admins ★

Ongoing Management

- Create Sub-Admins, assign departments, configure their permissions ★
- Institution-wide analytics: engagement, completion rates, accreditation reporting
- Override any Sub-Admin decision if escalation is needed ★

6A. SUB-ADMIN USER JOURNEY

★ **NEW** This section is new in v3.0. It documents the full end-to-end experience of the Sub-Admin role.

Phase 1: Activation (Day 0)

The Sub-Admin is created by the Tenant Admin. They receive a magic-link activation email branded to the institution.

Touchpoint: Activation Email

- Subject: "You've been made a Department Admin at [Institution Name]"
- Body: explains their role, which department they manage, link to activate account
- Clicks link → lands on branded password creation page → password created → logged in

First Login Experience

- Shown a brief 4-step orientation tour specific to the Sub-Admin dashboard (skippable)
- Tour covers: Instructor Overview Panel, Course Approval Queue, Department Analytics, Announcement Tool
- Prompted to review their assigned instructors list and verify it is correct — a "Confirm Department Roster" step

Phase 2: Setting Up The Department (Week 1)

Day 1–2: Review Assigned Instructors

- Open Instructor Overview Panel — see all assigned instructors with their profile cards
- Check each instructor's profile: courses listed, last login, any pending course submissions
- If an instructor is missing or wrongly assigned → contact Tenant Admin to reassign

Day 3–5: Process Pending Course Approvals

- Notification: "3 courses are pending your review"
- Open Course Approval Queue → review each course: syllabus, learning objectives, content, assessments
- Course 1: Approved — instructor notified, course published for student enrollment
- Course 2: Request Changes — Sub-Admin writes: "Please add learning objectives to Module 3 and clarify the assessment rubric" → instructor notified
- Course 3: Approved — published

Phase 3: Active Semester Management (Weeks 2–15)

Weekly Routine

- Monday: check department dashboard — attendance rates, quiz averages from last week
- Identify instructors with low engagement scores (e.g., only 60% of students attending live classes)
- Send an internal message to that instructor: "Noticed attendance has dropped in your Monday session — would you like to discuss?"
- Wednesday: course approval queue — new submissions from instructors for next module
- Review and approve within the same day to keep the course schedule on track

- Friday: at-risk student report — AI has flagged 5 students in Computer Science department with declining engagement
- Sub-Admin checks: which courses, which instructors are responsible, what the grade trend is
- Notifies each relevant instructor with context: "Student [ID] has missed 3 live classes and scored below 60% on last 2 quizzes"

Monthly: Instructor Performance Review

- Pull instructor performance report: sort by student satisfaction rating
- Identify top performer of the month → flag to Tenant Admin for recognition
- Identify lowest performer → review their course analytics, schedule a support check-in
- Export report as PDF → share with Tenant Admin at monthly review meeting

Phase 4: Semester End

Course Archiving

- End of semester: Sub-Admin reviews all active courses in department
- Archives completed courses: students retain read-only access, no new submissions accepted
- Instructors and students are automatically notified of archiving

Accreditation Contribution

- Generate department-level NAAC/NBA-compatible data export
- Includes: attendance rates, assessment scores, course completion rates, AI usage stats
- Export submitted to Tenant Admin who compiles the institution-wide accreditation report

Sub-Admin Emotional Journey

Stage	State	Platform Response
Day 1	Slightly uncertain about scope of authority	4-step orientation tour clarifies exactly what they can and cannot do
Week 1	Building familiarity with the dashboard	First course approval completed — quick win, confidence builds
Week 4	Comfortable with weekly routine	Dashboard analytics helping make data-driven decisions without needing to ask Tenant Admin
Week 8	Efficiently managing 10+ instructors	At-risk alerts preventing student failures; instructors appreciate the support check-ins
End of semester	Confident department manager	Accreditation export generated in minutes — tangible institutional value delivered

7. TECHNICAL REQUIREMENTS

7.1 Performance Requirements

Operation	Target (p95)	Maximum (p99)
Page Load (Dashboard, Course Pages)	<2 seconds	<4 seconds
Sub-Admin Instructor Dashboard Load ★	<3 seconds	<5 seconds
API Response (Read Operations)	<500 ms	<1 second
API Response (Write Operations)	<1 second	<2 seconds
Live Class Join	<5 seconds	<10 seconds
AI Query Response	<5 seconds	<10 seconds
Sub-Admin Analytics Export ★	<30 seconds (12 months)	<60 seconds
Search Results	<1 second	<2 seconds

7.2 Sub-Admin Data Model

★ **NEW** New database schema additions for the Sub-Admin role in v3.0.

Every Sub-Admin record in MongoDB contains the following structure:

Sub-Admin Document Schema

- `tenant_id`: ObjectId — mandatory; ties Sub-Admin to their institution
- `sub_admin_id`: ObjectId — unique identifier
- `user_id`: ObjectId — reference to the base user record (name, email, password hash)
- `assigned_departments`: [String] — list of department names this Sub-Admin manages
- `assigned_instructor_ids`: [ObjectId] — list of instructor user IDs under their supervision
- `permissions`: { `can_approve_courses`: Boolean, `can_send_announcements`: Boolean, `can_view_student_grades`: Boolean, `can_export_reports`: Boolean }
- `created_by`: ObjectId — Tenant Admin user ID who created this Sub-Admin
- `created_at`: DateTime — ISO 8601 UTC
- `status`: Enum [active, suspended, deleted]
- `last_login`: DateTime

All queries performed by or on behalf of a Sub-Admin include both `tenant_id` AND a department filter derived from `assigned_departments`. This is enforced at the middleware layer — the Sub-Admin cannot modify their own scope.

7.3 Security Requirements

Authentication & Session

- Password policy: 8+ chars, uppercase, lowercase, number, special char; lockout after 5 failed attempts
- JWT RS256 tokens; 24-hour expiry (Sub-Admin); refresh tokens valid 30 days
- Sub-Admin sessions expire after 60 minutes of inactivity
- MFA: optional per Sub-Admin account, configurable by Tenant Admin

Role-Based Access Control (RBAC)

Role	Created By	Manages
Super Admin	Mentora (internal)	All tenants, platform infrastructure
Tenant Admin	Super Admin (at provisioning)	Entire institution, all Sub-Admins, all users
Sub-Admin ★	Tenant Admin	Assigned departments only — instructors, courses, students in scope
Instructor	Tenant Admin or Sub-Admin (bulk import)	Their own courses only
Teaching Assistant	Instructor or Tenant Admin	Assigned courses — grading only
Student	Tenant Admin (bulk import)	Enrolled courses only
Parent (optional)	Tenant Admin	Read-only: their child's progress

7.4 Integration Requirements

SSO

- OAuth 2.0: Google, Microsoft, GitHub · SAML 2.0: Okta, OneLogin, Azure AD
- Sub-Admin accounts support SSO via the same provider as the institution

Student Information System (SIS)

- Supported: Ellucian Banner, Oracle PeopleSoft, Workday Student, Generic REST API
- Department assignments in SIS can auto-map to Sub-Admin scope via API field mapping ★

RESTful API

- Base URL: <https://api.mentora.com/v1/> · Bearer JWT or API key · 1,000 req/hour default
- New endpoints for Sub-Admin ★: GET /sub-admins, POST /sub-admins, GET /sub-admins/:id/instructors, GET /sub-admins/:id/courses, GET /sub-admins/:id/analytics
- Webhooks: sub_admin.created, course.submitted_for_approval, course.approved, course.rejected

8. SECURITY & COMPLIANCE

8.1 Data Residency and Sovereignty

- Global Cloud: nearest AWS region (Mumbai, Singapore, Frankfurt, Virginia, São Paulo)
- India-Sovereign: data never leaves Indian borders (AWS Mumbai + Pune)
- Private Cloud: customer on-premises deployment

8.2 Compliance Certifications Roadmap

Quarter	Certification	Status
Q1 2025	GDPR Compliance	Completed
Q2 2025	SOC 2 Type I	Completed
Q3 2025	SOC 2 Type II	In Progress
Q4 2025	ISO/IEC 27001:2022	Planned
2026	FERPA (US) + PIPEDA (Canada)	Planned

8.3 Sub-Admin Security Controls

★ **NEW** Specific security controls for the Sub-Admin role added in v3.0.

Sub-Admin Security Guarantees

- Database enforcement: every Sub-Admin query automatically includes tenant_id + department_id filter — cannot be bypassed through the API
- No privilege escalation: Sub-Admin cannot grant themselves additional permissions or create other Sub-Admins
- Immutable audit log: every Sub-Admin action is written to an append-only log; Sub-Admin cannot view or delete their own log
- Session isolation: Sub-Admin session token includes department_ids claim; backend validates this on every request
- Rate limiting: Sub-Admin API calls rate-limited to 500 requests/hour to prevent data harvesting
- Export watermarking: all PDF/CSV exports include Sub-Admin's name and timestamp — enabling traceability if data is shared externally

8.4 Data Retention Policies

Data Type	Retention Policy
User accounts (active)	Duration of enrollment + 1 year
Grades and transcripts	Permanent (legal requirement)
AI chat history	90 days (rolling deletion)
Sub-Admin audit logs ★	7 years (regulatory compliance)
Sub-Admin exported reports ★	2 years (downloadable copies tracked)

Inactive accounts (2+ years)	Personal data anonymized
Daily/Weekly/Monthly backups	7 days / 30 days / 1 year

8.5 Academic Integrity

- Turnitin API — checks internet, academic publications, past submissions
- GPTZero / Originality.AI for AI-generated content detection
- Online proctoring (optional): Live (30–50 INR), Recorded (15–25 INR), Automated AI (5–10 INR)

8.6 Incident Response SLAs

Priority	Acknowledgment	Resolution Target
P0 — Critical (platform down, breach)	15 minutes	Engineer assigned immediately
P1 — High (major feature unavailable)	1 hour	4 hours
P2 — Medium (non-critical feature)	4 hours	24 hours
P3 — Low (cosmetic, feature request)	24 hours	1 week

9. GO-TO-MARKET STRATEGY

9.1 Target Customer Segmentation

Segment	Profile	Key Value Proposition
Private Colleges (Primary)	500–5,000 students, Tier 2/3 cities	"Launch your digital campus in under an hour. Sub-Admin layer means your HODs manage their own departments — no IT team needed."
Corporate Training (Secondary)	5,000–50,000 employees; formal procurement	"Build your corporate university. Training managers become Sub-Admins, each owning their department's curriculum."
Coaching Institutes (Tertiary)	10,000–100,000 students; entrepreneurial buyers	"Clone your best teacher. Centre heads become Sub-Admins — each running their branch's courses independently."

9.2 Sales Strategy — Sub-Admin as a Sales Lever

How the Sub-Admin Feature Closes Deals

- Objection: "Our admin team is too small to manage 200 instructors." → Response: "Sub-Admins solve this — your HODs each manage their own department, you just oversee."
- Objection: "Our IT director doesn't want to be a bottleneck." → Response: "Sub-Admins take 90% of course approvals and instructor queries off the IT director's plate."
- Objection: "We have 5 campuses — who manages each?" → Response: "Create one Sub-Admin per campus. Each campus head manages their own instructors and courses independently."

7-Step Sales Process

Step	Key Activity
1 — Lead Generation	Inbound (SEO blog, webinars, 14-day free trial) + outbound (LinkedIn job-title targeting, cold calls, conferences)
2 — Discovery Call	30-min Zoom: understand pain, decision-makers, current tool stack, admin load distribution challenge
3 — Product Demo	60-min demo: show Sub-Admin creation, department assignment, instructor card overview, course approval flow ★
4 — Pilot Proposal	90-day free pilot; 1–2 departments; 1 Sub-Admin assigned; success criteria agreed upfront
5 — Pilot Execution	Weekly check-ins; CSM monitors Sub-Admin adoption; mid-pilot review at week 13
6 — Contract Negotiation	Pilot results to leadership; 1-year initial / 3-year preferred; DocuSign
7 — Expansion	Full rollout department by department; Sub-Admin per HOD; quarterly business reviews

9.3 Marketing Strategy

- Blog: 2 articles/week — target keywords: "best LMS India," "AI teaching assistant," "department admin LMS" ★

- Monthly webinar "Future of Learning" — guest speakers, 100–200 attendees, sales follow-up within 48 hours
- Case studies: detailed metrics-driven story for every customer — goal: 10 by Year 1 end
- LinkedIn Ads: 2,00,000 INR/month targeting Principal, Dean, HOD, CIO job titles ★
- Conference sponsorships: EdTechX Asia (5L INR), EDUCAUSE India (3L INR)

10. SUCCESS METRICS & KPIs

10.1 Platform Metrics

Metric	Year 1 Target	Year 3 Target
Active Institutions	100	500
Active Learners (MAU)	200,000	1.5 million
Sub-Admins Created ★	300 (avg. 3 per institution)	2,000+
Platform Uptime	99.9%	99.95%
Net Promoter Score	>50	>65
Annual Churn Rate	<5%	<3%
Annual Recurring Revenue	30 Crore INR	180 Crore INR

10.2 Sub-Admin Adoption Metrics

★ **NEW** New KPIs for the Sub-Admin role added in v3.0.

KPI	Target	Measurement Method
Sub-Admin adoption rate	>70% of institutions create at least 1 Sub-Admin within 30 days	Platform analytics — sub_admin.created events
Course approval turnaround	<48 hours from submission to approval/rejection decision	Avg. time between course.submitted and course.approved events
Tenant Admin time saved	>4 hours/week per institution (post Sub-Admin adoption)	Survey + action log comparison (before/after Sub-Admin creation)
Sub-Admin satisfaction	>4/5 stars in quarterly Sub-Admin experience survey	In-app survey sent to Sub-Admins every 90 days
Instructor satisfaction with Sub-Admin oversight	>3.8/5 stars	Anonymous instructor survey — "How useful is your department admin?"

10.3 Learning Outcome Metrics

- Students following AI study plans show average 12% grade improvement over 8 weeks
- AI tutoring resolves >82% of student doubts without human instructor involvement
- Average course completion rate >75% (industry average: 45%)
- Student satisfaction rating >4/5 on platform usefulness surveys

11. RISK ANALYSIS & MITIGATION

Risk	Probability	Impact	Mitigation
Large EdTech players pivot to B2B	Medium	High	First-mover advantage; focus on smaller institutions; superior Sub-Admin delegation model as differentiator
Sub-Admin scope creep (privilege escalation attempt)	Low	Critical	Database-level enforcement of department_id filter; automated security tests in CI/CD; immutable audit log; no self-modification of scope
Tenant Admin creates too many Sub-Admins, causing confusion	Medium	Medium	Platform recommends 1 Sub-Admin per department; onboarding guide includes best-practice ratios; Tenant Admin can deactivate quickly
AI model API costs escalate	Medium	Medium	Hybrid architecture: local Llama for routine queries, GPT-4o for complex; token budgets per tenant
Data breach or security incident	Low	Critical	Multi-layer security; annual pen testing; bug bounty; cyber insurance; incident response runbooks
Faculty resistance to Sub-Admin oversight	Medium	Medium	Instructor permissions unchanged — Sub-Admin is a support layer, not a performance enforcement layer; framing in onboarding
WebRTC infrastructure failure	Low	High	Auto-failover to Zoom/Teams/Google Meet; 99.95% Agora SLA

12. IMPLEMENTATION ROADMAP

Year 1 — Foundation

Quarter	Milestone	Success Criteria
Q1 2025	Platform MVP; first 10 pilot institutions; Tier 1–2–3 roles live	80% pilot-to-paid conversion; zero critical bugs
Q2 2025	AI Teaching Assistant GA; mobile app; SOC 2 Type I; Sub-Admin role MVP ★	Sub-Admin creation + instructor assignment + course approval working end-to-end
Q3 2025	Sub-Admin analytics dashboard GA ★; expansion to Maharashtra & Gujarat; 50 institutions	70% of institutions using Sub-Admin feature; NPS >50
Q4 2025	AI Grading GA; Cornell Notes GA; Sub-Admin API endpoints public ★; SOC 2 Type II	100 institutions; Sub-Admin adoption rate >70%; 30 Crore ARR

Year 2 — Scale

Quarter	Milestone
Q1 2026	Knowledge Graph; LTI integrations; Sub-Admin multi-department support ★
Q2 2026	Government institution tenders; ISO 27001; 200 institutions
Q3 2026	International expansion pilot (UAE, Saudi Arabia); FERPA compliance; 350 institutions
Q4 2026	Profitability milestone; Series A; 500 institutions; 180 Crore ARR

13. APPENDICES

Appendix A — Glossary

Term	Definition
Tenant	An organization (college, company, coaching institute) that subscribes to Mentora and gets its own isolated workspace
Sub-Admin ★	A department-level administrator created by the Tenant Admin; manages assigned instructors and courses within a scoped department
HOD	Head of Department — the typical real-world person who would hold the Sub-Admin role in an academic institution
LMS	Learning Management System — software for administering, documenting, and delivering educational courses
RAG	Retrieval-Augmented Generation — AI technique combining semantic search with language model generation
WebRTC	Web Real-Time Communication — open standard for real-time audio/video in browsers (used for live classes via Agora.io)
NAAC	National Assessment and Accreditation Council — India's higher education accreditation body
NBA	National Board of Accreditation — India's technical program accreditation body
JIT Provisioning	Just-In-Time provisioning — auto-creating user accounts on first SSO login
BANT	Budget, Authority, Need, Timeline — sales lead qualification framework
ARR	Annual Recurring Revenue — annualized value of subscription contracts
NPS	Net Promoter Score — customer loyalty metric (–100 to +100)
department_id	The database field used to scope all Sub-Admin queries; always combined with tenant_id in compound indexes

Appendix B — Document Information

Document Version	3.0
Last Updated	May 2025
Document Owner	Product Team
Status	Final – Awaiting Approval
Classification	Confidential — Internal Use Only
Key Change in v3.0	Sub-Admin (Department Admin) role — Sections 4A, 5A, 6A, 7.2, 8.3, 10.2 are new or updated